P-A-0086b(DSA-DM)
November 17, 1970
Interim Revision of
Fed Spec P-A-86a
August 14, 1957

INTERIM FEDERAL SPECIFICATION

ABRASIVE PASTE, DENTAL

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification P-A-86a.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers dental abrasive pastes, suitable as abrading and polishing agents for plastic and porcelain teeth, and for cast metal restorations.

1.2 Classification. Dental abrasive pastes covered by this specification shall be of the following types, as specified (see 6.1):

Type I. Silicon carbide paste.
Type II. Aluminum oxide paste.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FSC 6520

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SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.
PPP-C-00186a	Container, Packaging and Packing For Drugs, Chemicals and Pharmaceuticals.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which included cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Auburn, Washington, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

SPECIFICATION

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

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(Copies of Military specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN PHARMACEUTICAL ASSOCIATION

National Formulary

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042).

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The Pharmacopeia of the United States of America

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042).

U. S. DEPARTMENT OF COMMERCE - National Bureau of Standards

Commercial Standard CS 271-65 Grading of Abrasive Grain for Grinding Wheels.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402).

3. REQUIREMENTS

3.1 Materials. Materials shall be of the highest quality throughout and suitable in all respects for the intended purpose. Any ingredients entering into the paste which are not included in the Pharmacopeia of the United States of America (U.S.P.) or National Formulary (N.F.) shall be of quality suitable for pharmaceutical use.

3.2 Composition.

3.2.1 Type I, silicon carbide paste. Type I paste shall consist essentially of silicon carbide incorporated in an aqueous gel, and shall contain not less than 60 percent by weight of silicon carbide. The gel shall be a vegetable colloid in glycerin with modifiers, preservatives, color and flavor additives. The paste shall contain no harmful ingredients, shall be free from greases and oils, shall be neutral or alkaline in reaction, and shall be pleasant to the taste. Color of paste shall be black.

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3.2.2 Type II, aluminum oxide paste. Type II paste shall consist essentially of spherically shaped aluminum oxide and zinc oxide, incorporated in a nonwater soluble base such as petrolatum. The base shall contain modifiers, preservatives (for non-mineral bases), color and flavor additives. The paste shall contain no harmful ingredients, and be pleasant to the taste. Color of paste shall be a light shade of gray.

3.3 Description.

3.3.1 Type I, silicon carbide paste. Silicon carbide paste shall be suitable for intra- or extra-oral grinding of porcelain teeth. Paste shall be removable by washing with warm water.

3.3.2 Type II, aluminum oxide paste. Aluminum oxide paste shall be an all purpose paste for intra- or extra-oral milling, finishing, and polishing of plastic teeth, and for cast metal restorations. Paste shall be removable by brushing with soap and warm water.

3.4 Particle size. Particle size of silicon carbide and aluminum oxide shall be 150 mesh, when tested as specified in 4.4.2.

3.5 Consistency.

3.5.1 Type I, silicon carbide paste. The average of the major and minor diameters of the slumped mass of paste shall be 27.0 mm. $\pm$ 2.0 mm., when tested as specified in 4.4.3.1.

3.5.2 Type II, aluminum oxide paste. The paste shall not flow or spread more than 1/64 inch under compression, when tested as specified in 4.4.3.2.

3.6 Net weight. The net weight of Types I and II pastes shall be 3.0 oz. minimum (85 grams min.), when tested as specified in 4.4.4.

3.7 Stability. The paste shall conform to the requirements of this specification for no less than two years from date of delivery to the Government, when stored in a cool dry place (50° F. to 80° F.).

3.8 Delivery schedule. No more than three months shall have elapsed from date of manufacture of the paste to date of delivery to the Government.

3.9 Manufacturer's instructions. Adequate and accurate instructions for specific uses and technique for use of each abrasive paste, shall be furnished with each unit of issue. In addition, two copies of the instructions shall be forwarded to the contracting officer for information purposes.

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3.10 Workmanship. Workmanship shall be first class throughout. The product and its containers shall be free from defects which detract from its appearance or impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the raw material, may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in Tables I and III (see 4.2.1 and 4.3) and Table II (see 4.2.2).

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4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in Table I. The unit of product for examination purposes shall be one tube of paste.

Table I. Examination Criteria

Type defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with table II. The unit of product for test purposes shall be one tube of paste.

Table II. Tests

Characteristic	Reqmt.	Test Procedure	Unit of Product to be Tested per lot or batch number	Number of determinations per unit of product	Total number of determinations
Identification	3.2	4.4.1	See 4.2.3	1 a/	1
Particle Size	3.4	4.4.2	See 4.2.3	1 a/	1
Consistency					
Type I	3.5.1	4.4.3.1	1	3 b/	3
Type II	3.5.2	4.4.3.2	1	1 a/	1
Net Weight	3.6	4.4.4	1	1 a/	1

a/ A lot shall be rejected if any determination fails the test requirement.

b/ A lot shall be rejected if the average of the determinations per unit of product fails the test.

4.2.3 Sampling of raw materials. A sufficient quantity of each lot of silicon carbide or aluminum oxide, as applicable, used in the formulation of the abrasive paste shall be selected for identification and particle size.

4.2.4 Sample unit usage. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

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4.3 Examination. The tube of paste shall be examined for defects including, but not limited to, those listed in table III.

Table III. Classification of Defects

Categories	Defects
Critical:	None defined.
Major:	
101	Type of abrasive paste not as specified.
102	Paste not free from foreign matter, dirt, or other impurities.
103	Paste not uniform, smooth, or homogeneous product.
104	Paste will not extrude from tube without excess pressure.
105	Tube shows leakage.
Minor:	
201	Instructions for use are inadequate, incomplete, inaccurate, or missing altogether.
202	Tube deformed.

4.4 Tests.

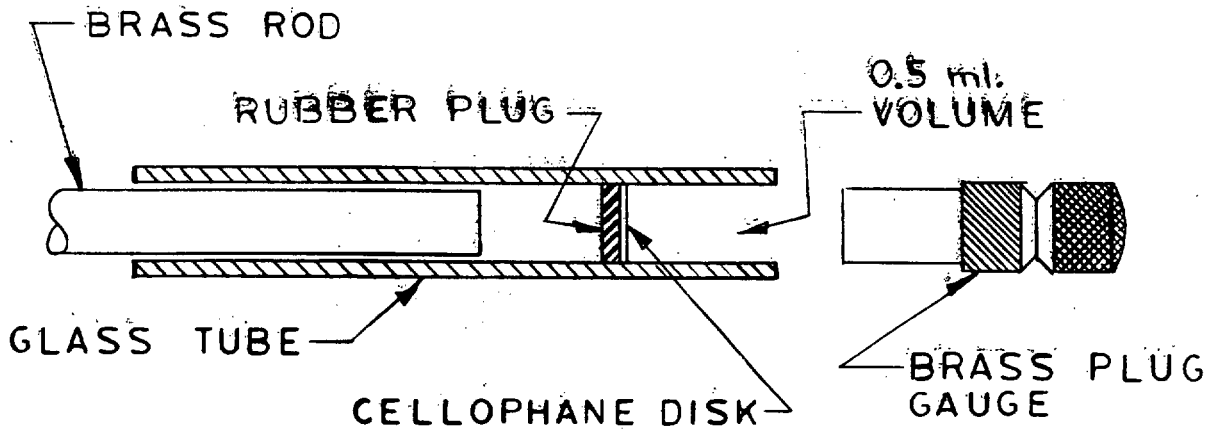
4.4.1 Identification. In the absence of certificates of quality for the silicon carbide or aluminum oxide, identification shall be conducted in accordance with acceptable qualitative analytical methods.

4.4.2 Particle size. Particle size of silicon carbide and aluminum oxide shall be determined in accordance with Table 1 of U.S. Department of Commerce Commercial Standard CS 271-65.

4.4.3 Consistency. Consistency tests shall be conducted at a temperature of 23.0 ± 2.0° C. and a relative humidity of 5 0 ± 10 percent. Materials and test equipment shall be conditioned in the test room for not less than ten hours prior to testing.

4.4.3.1 Type I, silicon carbide paste. The type of apparatus for measuring consistency shall be as shown in Figure 1. The apparatus consists of two flat glass plates, a weight, a gauge plug, and a glass tube (inside diameter approximately 6.5 mm.) which will deliver 0.50 ml. of abrasive paste. The combined weight of the top plate and the weight shall be 120 grams. The paste (0.50 ± 0.02 ml.) shall be delivered from the glass tube onto a flat glass plate. The top glass plate, weighing approximately 20 grams, and the necessary additional weight required to total 120 grams shall be carefully lowered onto the paste. The average of the major and minor diameters of the slumped mass of paste shall be determined as soon as there is no visual dimensional change. The average of three such determinations shall be the consistency of the silicon carbide abrasive paste.

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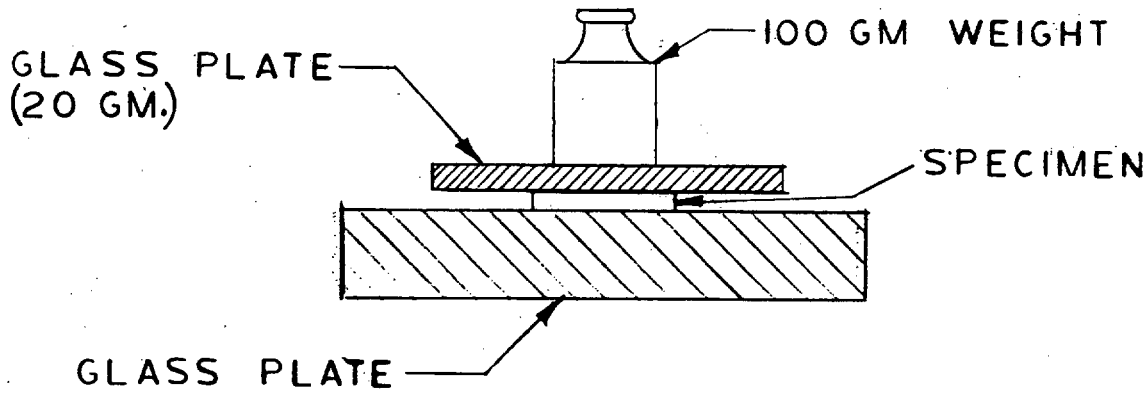


FIGURE -1-

APPARATUS FOR CONSISTENCY TESTS

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4.4.3.2 Type II, aluminum oxide paste. Aluminum oxide paste (0.60 ± 0.02 ml.) shall be dispensed onto a glass plate with a level surface. Place a "silver" quarter on top of the paste, and compress the paste until it spreads out to the circumference edge of the coin. After 5 minutes observe if paste has spread more than 1/64 inch beyond the coin.

4.4.4 Net weight. The net weight of the paste shall be determined using a balance suitable for the intended purpose.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Unit of issue. Three ounces of abrasive paste, of the type specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table IV, shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Type	Unit package	Intermediate package	Exterior container
1 or 2	1 unit	6 units	96 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Immediate container. Three ounces of aluminum oxide or silicon carbide shall be packaged in a collapsible metal tube, conforming to PPP-C-00186a, group B, class 1, type a, closure A.

5.1.3.2 Unit package. Unit package shall be a box of appropriate size, constructed in accordance with PPP-B-566 or PPP-B-676 except that commercial colors are acceptable. Closure shall be adequate to prevent spilling of contents under normal handling.

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5.1.3.3 Intermediate package. Intermediate package of appropriate size shall be constructed in accordance with PPP-B-566, PPP-B-676 or PPP-B-636, type CF, class domestic. Commercial colors are acceptable on intermediate packages. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each level A box shall be lined with a waterproof case liner conforming to Military Specification MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed in accordance with 30.4 of PPP-B-636.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure of wood boxes shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II as specified in the appendix of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.3 Marking.

5.3.1 Immediate container. In addition to any commercial markings, each tube shall be permanently and legibly marked to permit identification of abrasive paste. In addition, marking shall include the manufacturer's name, and address.

5.3.2 Interior packages . Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Lot (control) number, contract or purchase order number and name of contractor shall be shown. The date of manufacture shall be shown in lieu of the date packed. Marking shall include the legend "STORE IN A COOL PLACE (50°-80° F.)." In addition, instructions as specified in 3.9, shall be placed in each unit package.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control) number, the date of manufacture in lieu of the date packed, and the legend "STORE IN A COOL PLACE," shall be shown.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Specification changes. Because of the extensiveness of changes to this document, the margins have not been marked with an asterisk to indicate where changes from the previous issue have been made.

6.3 Limitations. This specification does not cover all types, classes, grades, styles, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover those items which are normally procured to meet Federal Government requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6520-1272

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements .

SPECIFICATION
P-A-0086b(DSA-DM) FSN:

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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